0.8A STEP-DOWN/STEP-UP/INVERTING DC-DC CONVERTER AZ34063

General Description

The AZ34063 is a monolithic switching regulator control circuit containing the primary functions required for DC-DC converters. This device consists of internal temperature compensated reference, voltage comparator, controlled duty cycle oscillator with active current limit circuit, driver and high current output switch. This device is specifically designed to be used in Step-Down, Step-Up and Voltage-Inverting applications with a minimum number of external components.

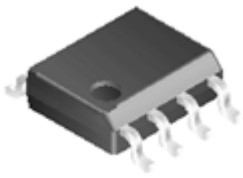
The AZ34063 is available in 2 packages: SOIC-8 and DIP-8.

Features

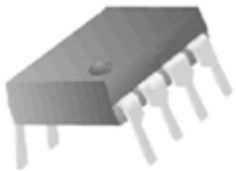
- Operation from 3.0V to 36V Input
- Low Standby Current
- Current Limiting
- Output Switch Current to 0.8A
- Output Voltage Adjustable
- Operation Frequency up to 65KHz (CT=330pF)
- Precision 2% Reference

Applications

- Battery Chargers
- NICs/ Switches/ Hubs
- ADSL Modems
- Negative Voltage Power Supplies



SOIC-8



DIP-8

Figure 1. Package Types of AZ34063

Pin Configuration

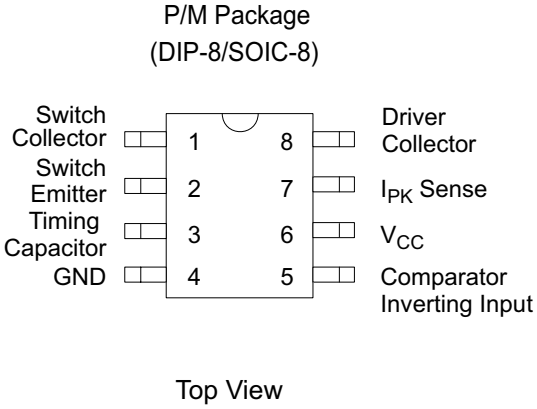


Figure 2. Pin Configuration of AZ34063

Functional Block Diagram

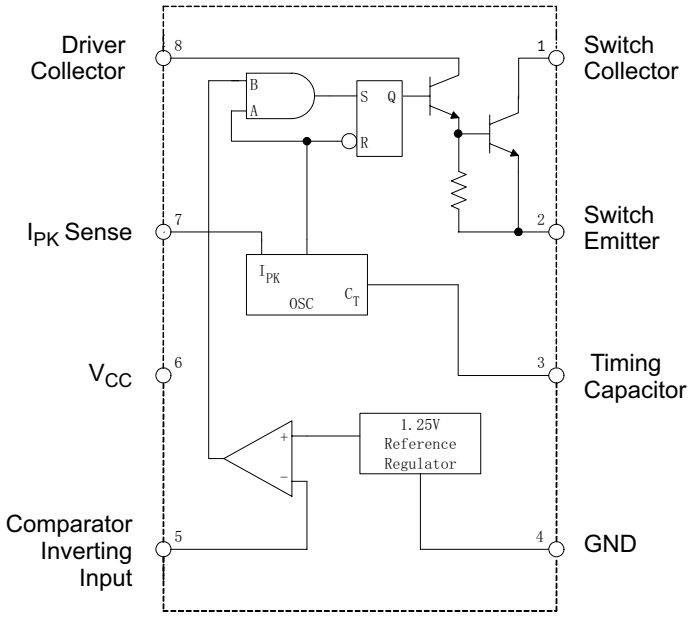
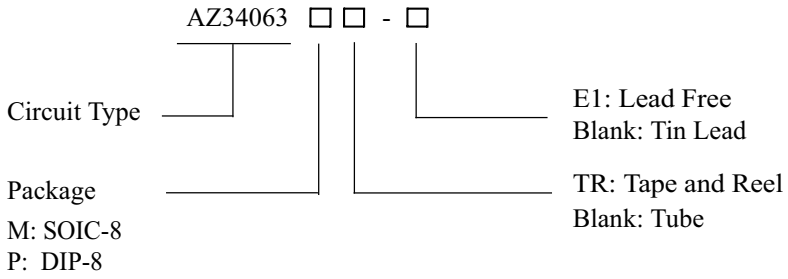


Figure 3. Functional Block Diagram of AZ34063

Pin Description

Pin Number	Pin Name	Function
1	Switch Collector	Internal switch transistor collector
2	Switch Emitter	Internal switch transistor emitter
3	Timing Capacitor	Timing Capacitor to control the switching frequency
4	GND	Ground pin for all internal circuits
5	Comparator Inverting Input	Inverting input pin for internal comparator
6	V_{CC}	Voltage supply
7	I_{PK} Sense	Peak Current Sense Input by monitoring the voltage drop across an external I sense resistor to limit the peak current through the switch
8	Driver Collector	Voltage driver collector

Ordering Information



Package	Temperature Range	Part Number		Marking ID		Packing Type
		Tin Lead	Lead Free	Tin Lead	Lead Free	
SOIC-8	-40 to 85°C	AZ34063M	AZ34063M-E1	34063M	34063M-E1	Tube
		AZ34063MTR	AZ34063MTR-E1	34063M	34063M-E1	Tape & Reel
DIP-8	-40 to 85°C	AZ34063P	AZ34063P-E1	AZ34063P	AZ34063P-E1	Tube

BCD Semiconductor's Pb-free products, as designated with "E1" suffix in the part number, are RoHS compliant.

Absolute Maximum Ratings (Note 1)

Parameter	Symbol	Value	Unit
Power Supply Voltage	V_{CC}	40	V
Comparator Input Voltage Range	V_{IR}	-0.3 to 7	V
Switch Collector Voltage	V_C(switch)	40	V
Switch Emitter Voltage (Vpin1=40V)	V_E(switch)	40	V
Switch Collector to Emitter Voltage	V_{CE}(switch)	40	V
Driver Collector Voltage	V_C(driver)	40	V
Driver Collector Current (Note 2)	I_C(driver)	100	mA
Switch Current	I_{SW}	0.8	A
Power Dissipation and Thermal Characteristics Plastic Package, Power Dissipation (T_A = 25 ℃) Thermal Resistance SOIC Package, Power Dissipation (T_A = 25 ℃) Thermal Resistance	 P_D $R_{\theta J}$ P_D $R_{\theta JA}$	 1.25 100 625 160	 W ℃ /W mW ℃ /W
Operating Junction Temperature	T_J	150	℃
Storage Temperature Range	T_{STG}	-65 to 150	℃
ESD (Human body model)		2000	V

Note 1: Stresses greater than those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under "Recommended Operating Conditions" is not implied. Exposure to "Absolute Maximum Ratings" for extended periods may affect device reliability.
Note 2: Maximum package power dissipation limits must be observed.

Recommended Operating Conditions

Parameter	Symbol	Min	Max	Unit
Supply Voltage	V_{CC}	3	36	V
Ambient Temperature	T_A	-40	85	^oC

Electrical Characteristics

(V_{CC} = 5.0 V, T_A = -40 to 85°C, unless otherwise specified.)

Parameter	Symbol	Conditions		Min	Typ	Max	Unit
OSCILLATOR							
Frequency	f_{OSC}	VPin 5 = 0 V, T_A = 25°C	CT = 1.0 nF	24	33	42	KHz
			CT = 330 pF	50	65	80	KHz
Charge Current	I_{CHG}	V_{CC} = 5.0 V to 36 V, T_A = 25°C		24	35	42	µA
Discharge Current	I_{DISCHG}	V_{CC} = 5.0 V to 36 V, T_A = 25°C		140	220	260	µA
Discharge to Charge Current Ratio	I_{DISCHG}/I_{CHG}	Pin 7 to V_{CC}, T_A = 25°C		5.2	6.5	7.5	
Current Limit Sense	V_{IPK}(sense)	I_{CHG} = I_{DISCHG}, T_A = 25°C		250	300	350	mV
OUTPUT SWITCH (Note 3)							
Saturation Voltage, Dalington Connection	V_{CE}(sat)	I_{SW} = 0.6 A, Pins 1, 8 connected			1.0	1.3	V
Saturation Voltage (Note 4)	V_{CE}(sat)	I_{SW} = 0.6 A, RPin 8 = 82 Ω to V_{CC}, Forced β = 20			0.45	0.8	V
DC Current Gain	h_{FE}	I_{SW} = 0.6 A, V_{CE} = 5.0 V,		50	75		
Collector Off-State Current	I_C(off)	V_{CE} = 36 V			0.01	100	µA
COMPARATOR							
Threshold Voltage	V_{TH}	T_A = 25°C		1.225	1.250	1.275	V
		T_A = -40 to 85°C		1.21		1.29	
Threshold Voltage Line Regulation	R_{EGLINE}	V_{CC} = 3.0 V to 36 V			1.4	5	mV
Input Bias Current	I_{IB}	V_{IN} = 0 V			-20	-400	nA
TOTAL DEVICE							
Supply Current	I_{CC}	V_{CC} = 5.0 V to 36 V, CT = 1.0 nF, Pin 7 = V_{CC}, VPin 5 > V_{TH}, Pin 2 = GND, other pins open				4	mA

Note 3: Low duty cycle pulse technique are used during test to maintain junction temperature as close to ambient temperature as possible.

Note 4: If the output switch is driven into hard saturation (non-Darlington configuration) at low switch currents (≤ 300mA) and high driver currents (≥ 30mA), it may take up to 2.0us for it to come out of saturation. This condition will shorten the off time at frequencies 30KHz, and is magnified at high temperatures. This condition does not occur with a Darlington configuration, since the output switch cannot saturate. If a non-Darlington configuration is used, the following output drive condition is recommended:

Electrical Characteristics (Continued)

Forced β of output switch: $\frac{I_C \text{ output}}{I_C \text{ driver - } 7.0\text{mA}^*} \geq 10$

* The 100Ω resistor in the emitter of the driver device requires about 7.0 mA before the output switch conducts.

Typical Performance Characteristics

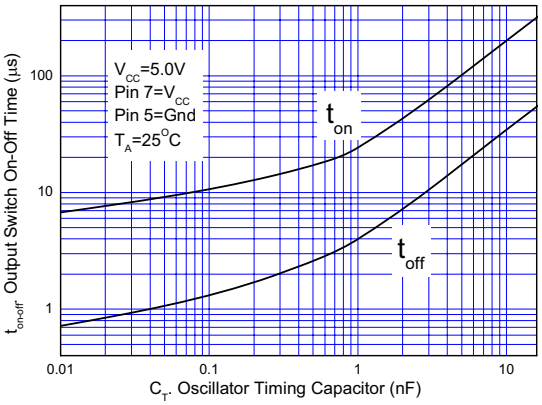


Figure 4. Output Switch On-Off Time vs. Oscillator Timing Capacitor

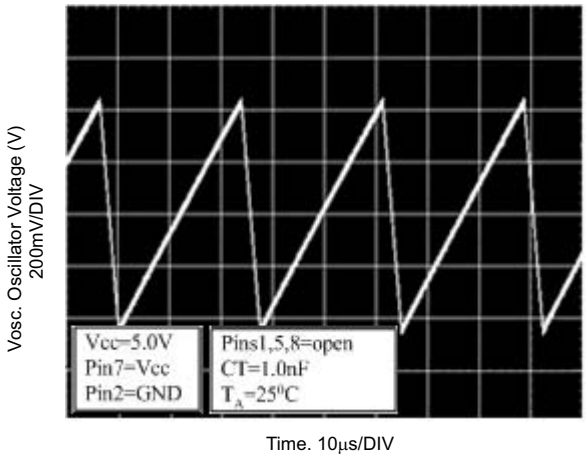


Figure 5. Timing Capacitor Waveform

Typical Performance Characteristics (Continued)

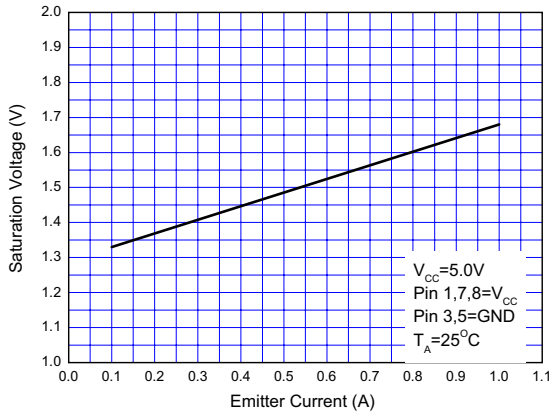


Figure 6. Emitter Follower Configuration Output Saturation Voltage vs. Emitter Current

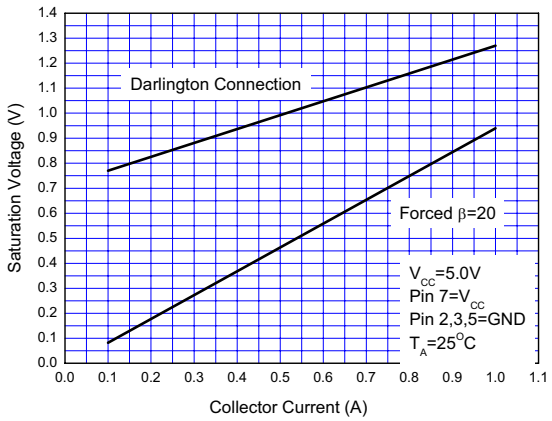


Figure 7. Common Emitter Configuration Output Switch Saturation Voltage vs. Collector Current

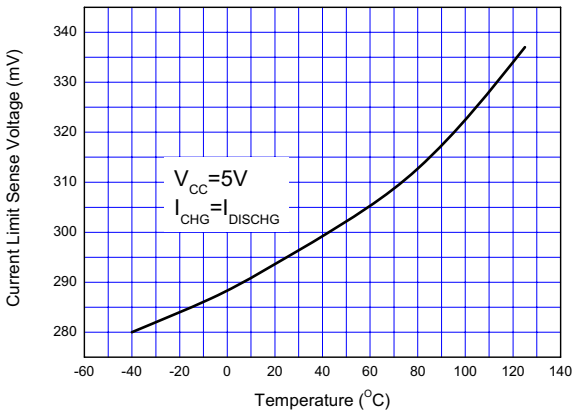


Figure 8. Current Limit Sense Voltage vs. Temperature

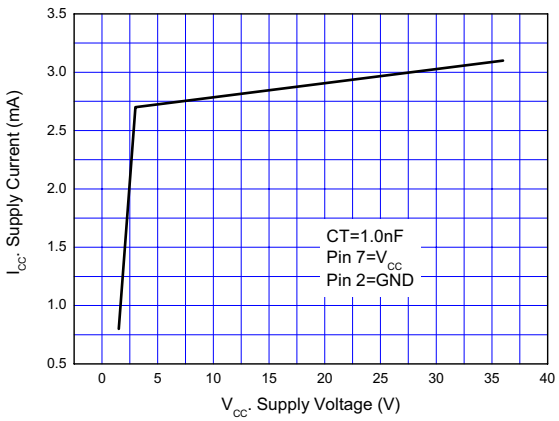


Figure 9. Standby Supply Current vs. Supply Voltage

Typical Applications

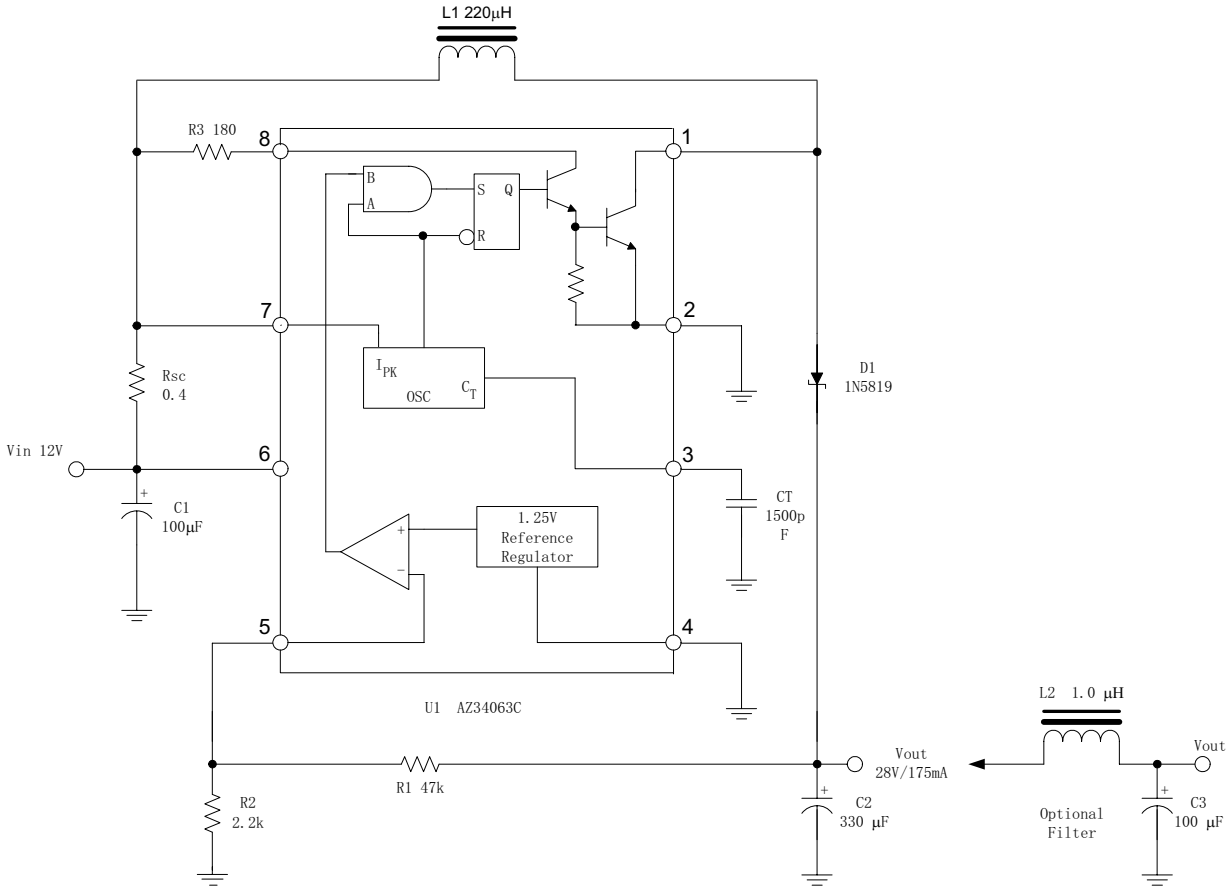


Figure 10. Step-Up Converter (Note 5)

Note 5: This is a typical step-up converter configuration. In the steady state, if the resistor divider voltage at pin 5 is greater than the voltage in the non-inverting input, which is 1.25V determined by the internal reference, the output of the comparator will go low. At the next swithching period, the output switch will not conduct and the output voltage will eventually drop below its nominal voltage until the divider voltage at pin 5 is lower than 1.25V. Then the output of the comparator will go high, the output switch will be allowed to conduct. Since Vpin5 = V_{OUT} * R2/(R1+R2) = 1.25(V), the output voltage can be decided by V_{OUT} = 1.25 * (R1+R2)/R2 (V).

Typical Applications (Continued)

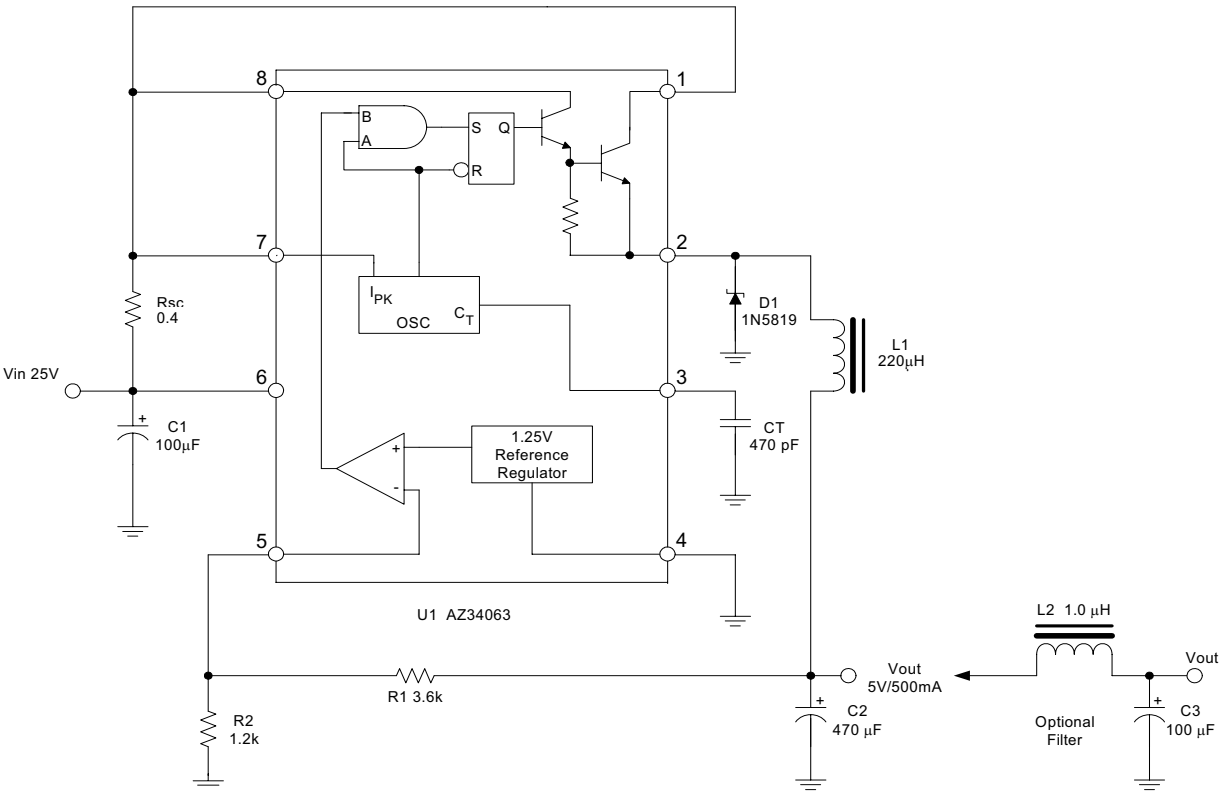


Figure 11. Step-Down converter (Note 6)

Note 6: This is a typical step-down converter configuration. The working process in the steady state is similar to step-up converter, Vpin5 = V_{OUT} * R2/(R1+R2) = 1.25 (V), the output voltage can be decided by V_{OUT} = 1.25 * (R1+R2)/R2 (V).

Typical Applications (Continued)

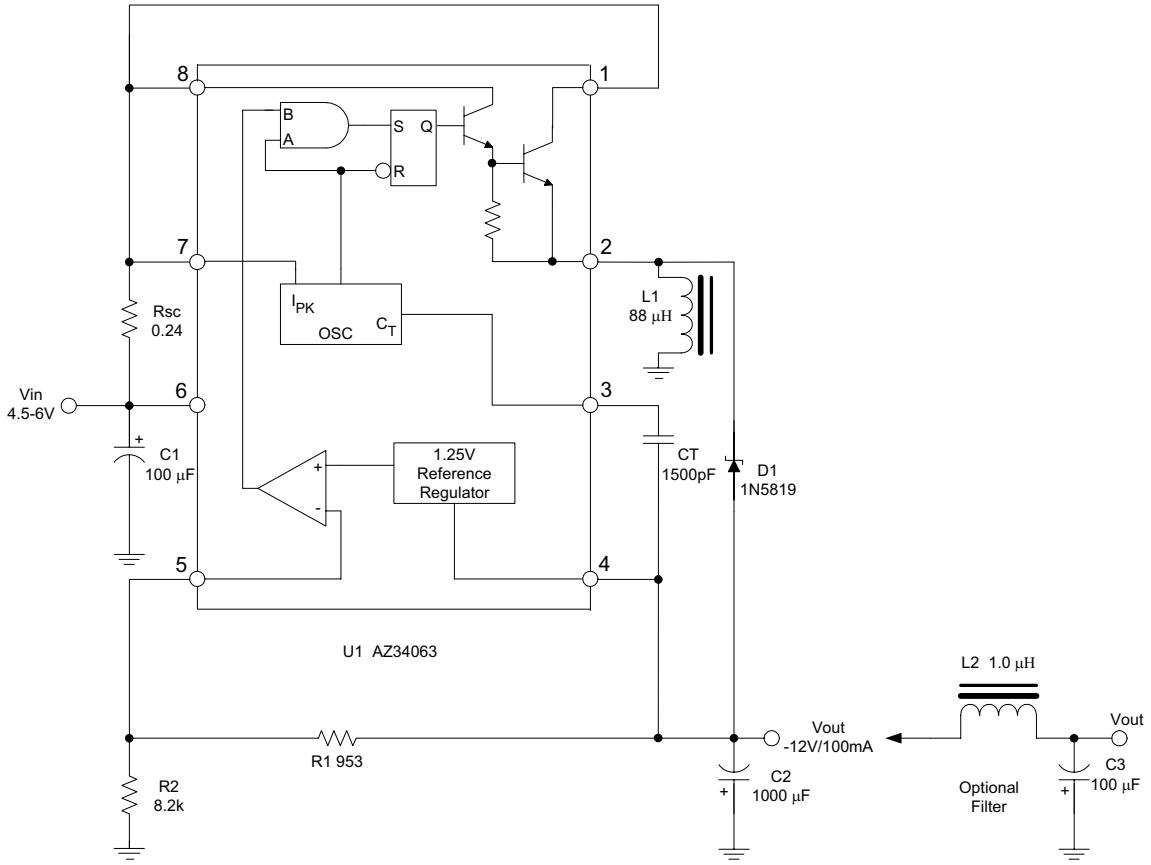
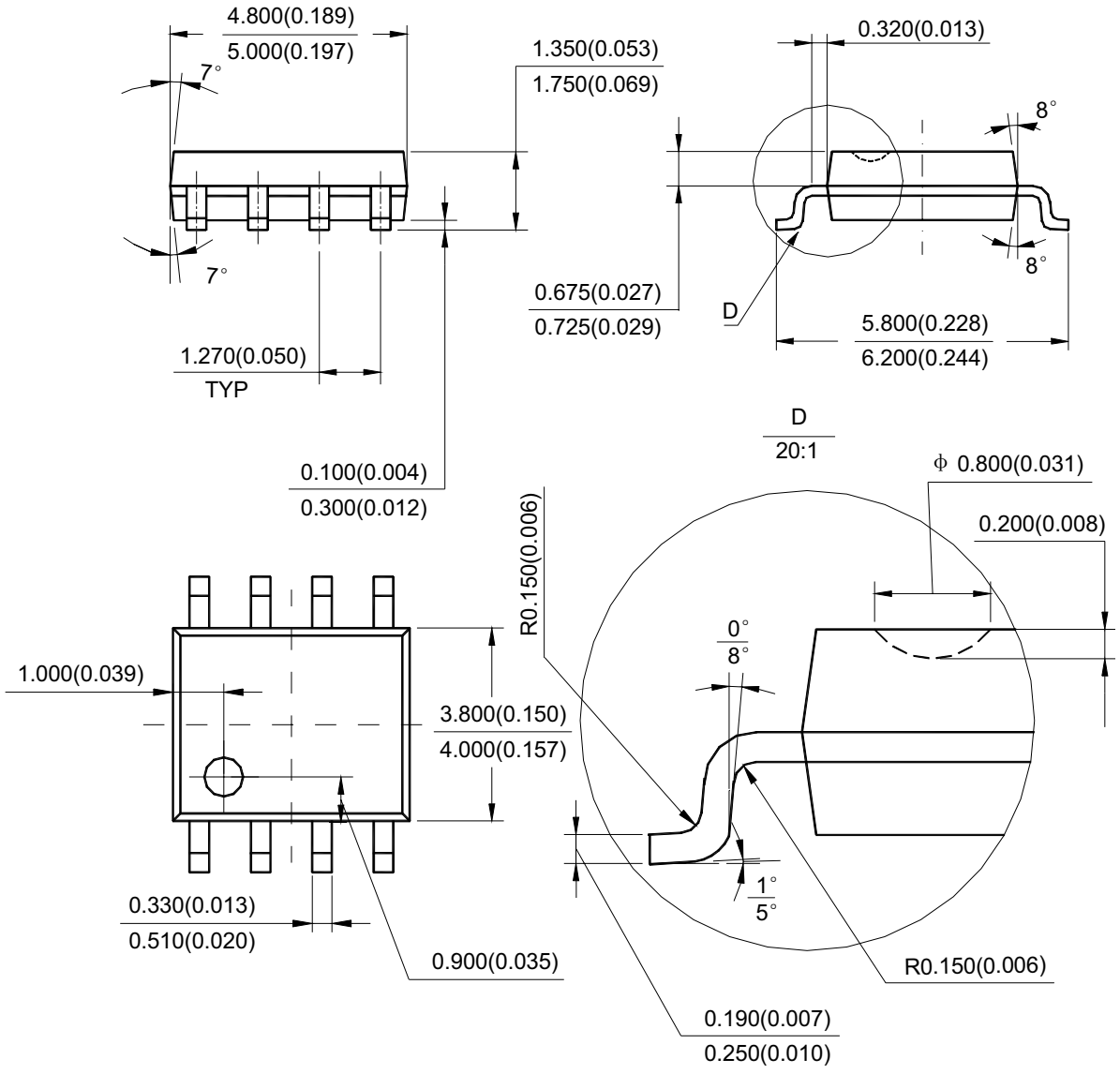


Figure 12. Voltage Inverting Converter (Note 7)

Note 7: This is a typical inverting converter configuration. The working process in the steady state is similar to step-up converter, the difference in this situation is that the voltage at the non-inverting pin of the comparator is equal to 1.25V+V_{OUT}, then Vpin5 = V_{OUT} * R2/(R1+R2) = 1.25V+V_{OUT}, so the output voltage can be decided by V_{OUT} = -1.25 * (R1+R2)/R1 (V).

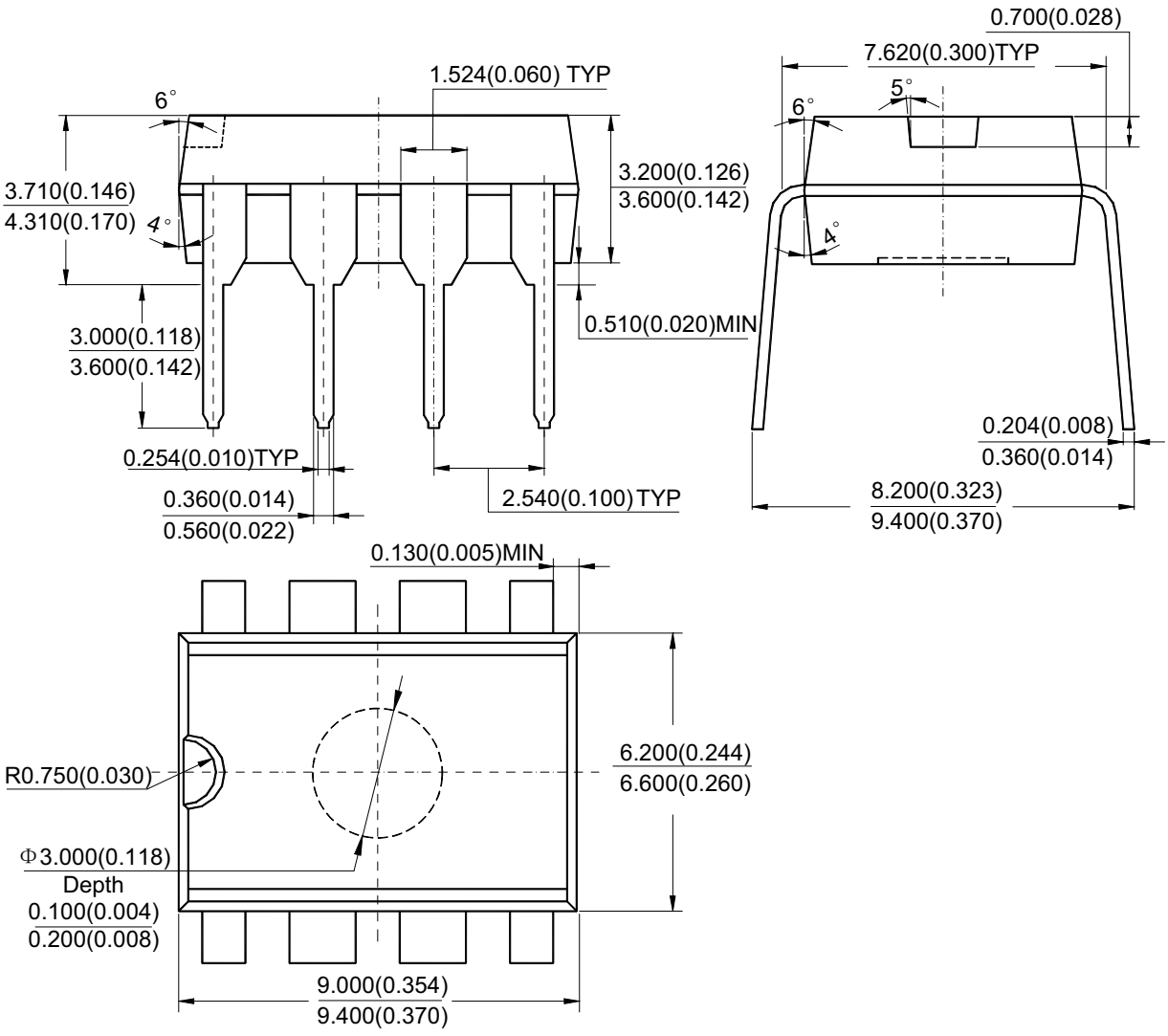
Mechanical Dimensions

SOIC-8 **Unit: mm(inch)**



Mechanical Dimensions (Continued)

DIP-8 **Unit: mm(inch)**





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